Instagram Engagement Prediction

1. Problem Statement

In today's digital landscape, Instagram engagement is a key indicator of content performance and audience interaction. Businesses and content creators rely on engagement metrics to optimize their strategies and maximize reach.

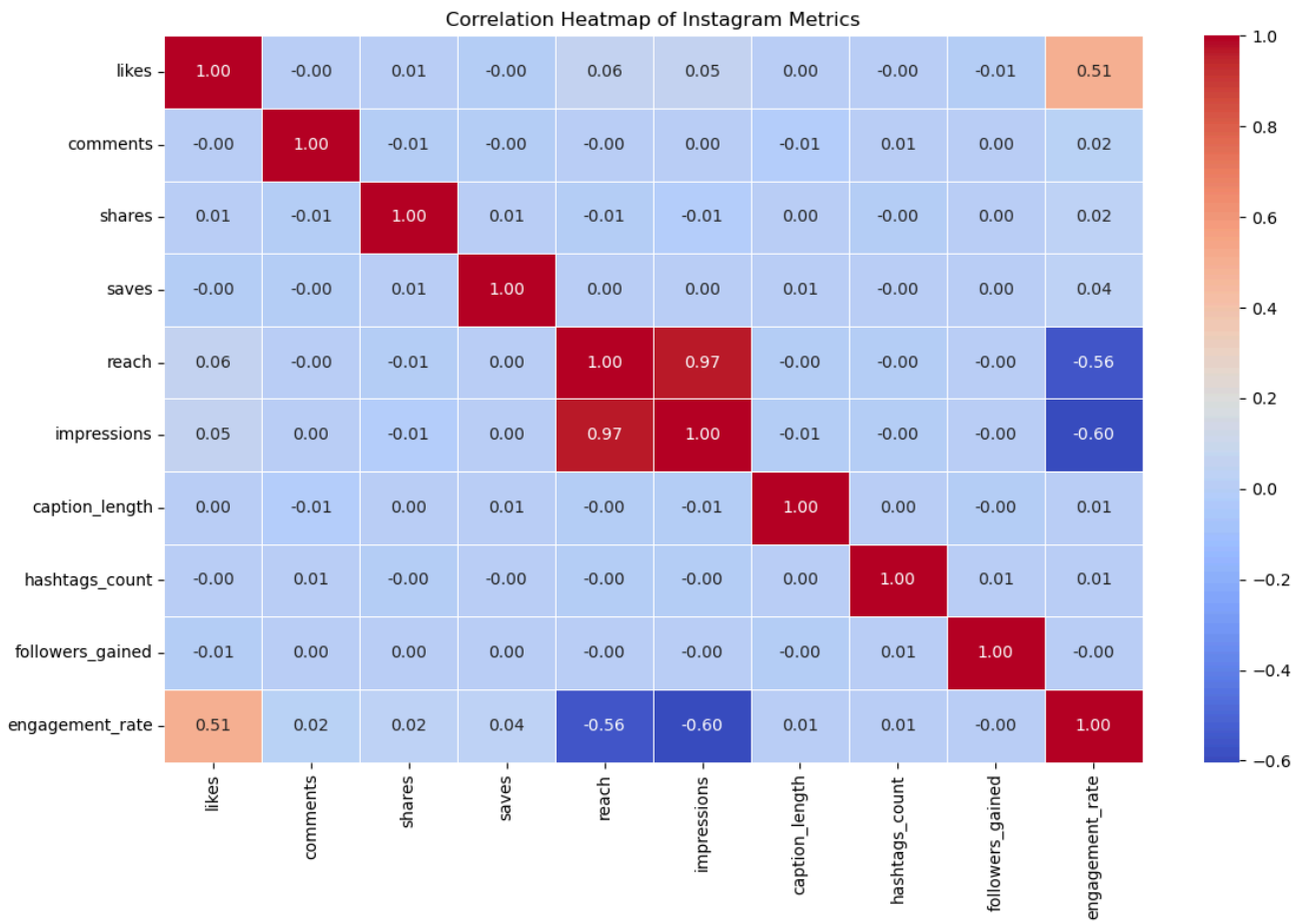
The objective of this project is to predict the engagement rate of Instagram posts using available metrics such as likes, comments, shares, saves and reach. By building predictive models, this project aims to identify the factors that most strongly influence engagement and provide actionable insights.

2. Data Overview

The dataset used in the project contains Instagram post-level analytics, including: likes, comments, shares, saves, reach and additional variables.

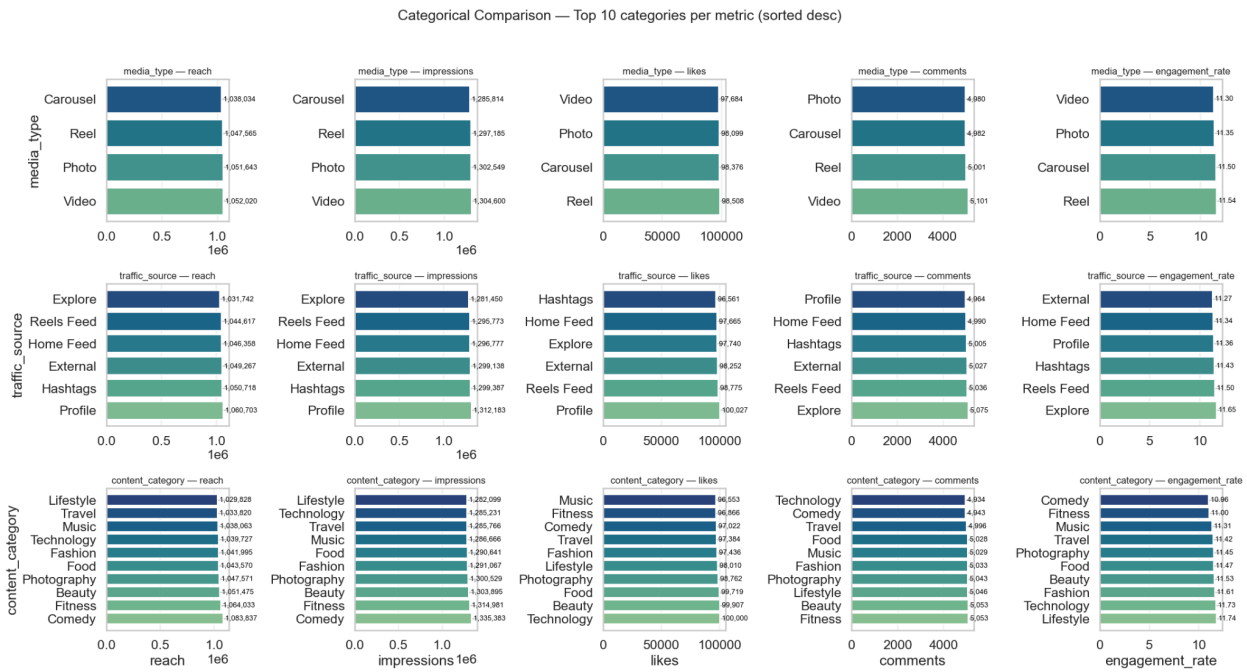
3. Exploratory Data Analysis

Understanding relationships between variables is critical before modeling.



Interpretation:

- Engagement rate shows a **moderate positive correlation (~0.51)** with key interaction metrics.
- Most other variables show weak correlations, suggesting limited independent predictive power.
- This indicates that engagement is primarily driven by interaction-based features.



Interpretation:

- The **Home Feed** is the dominant traffic source.
- Other sources like Explore and External contribute less consistently.
- This suggests that **content visibility is heavily platform-driven**, not evenly distributed.

4. Methodology

- Data Preprocessing
    - Loaded dataset and inspected structure
    - Handled missing values
    - Converted data types where necessary
    - Created the engagement rate feature using the defined formula
    - Selected relevant features for modeling

- Model Implemented

  Three models were developed and evaluated:

- Random Forest Regressor
- Linear Regression
- K-Nearest Neighbors (KNN)

- Evaluation Metrics

  - The models were evaluated using:
  - RMSE (Root Mean Squared Error)
  - MAE (Mean Absolute Error)
  - R^2 Score

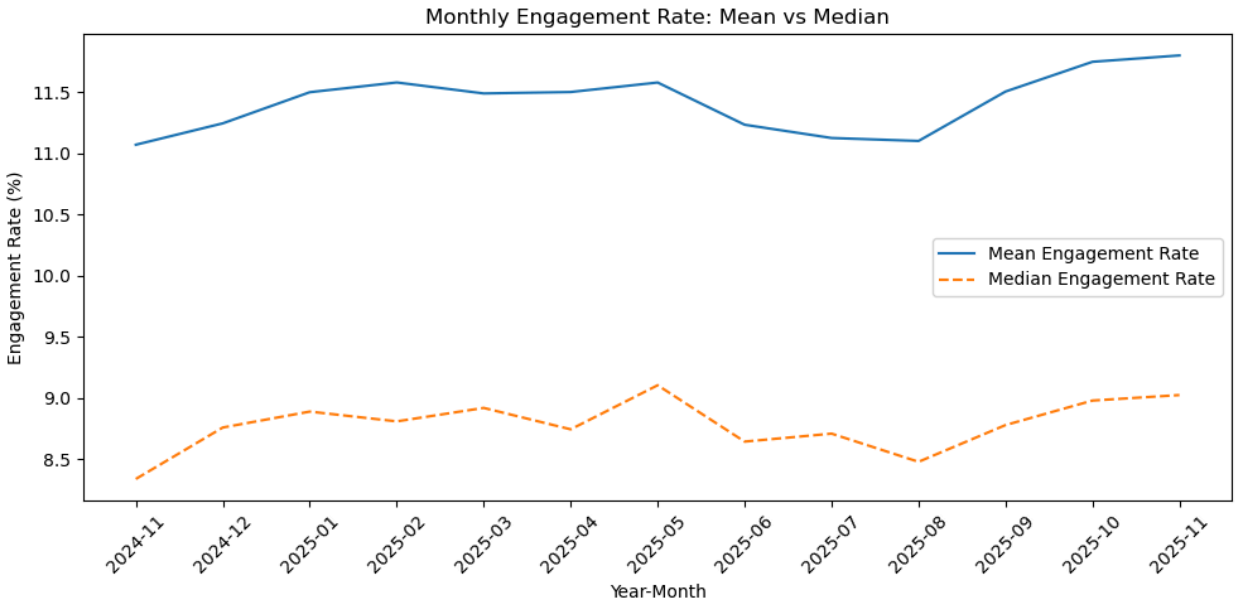
5. Model Performance

Models	**RMSE**	**MAE**	**R2**
Random Forest	0.7439	0.2983	0.9946
Linear Regression	5.9085	3.6592	0.6623
KNN Regressor	6.8589	3.6267	0.5450

- Interpretation

- The Random Forest model significantly outperformed the others, achieving a very high r2 score, indicating excellent predictive performance
- Linear Regression performed moderately, suggesting some linear relationship but not capturing complex interactions.
- KNN performance depends on tuning but generally falls between the two.

6. Behavior and Trends



Interpretation:

- Engagement rate fluctuates within a relatively narrow range.
- There are periodic increases, suggesting **content timing or type influences engagement**.
- No extreme volatility indicates stable audience behavior.

7. Critical Analysis

While the Random Forest model shows extremely high performance, this result must be interpreted with caution.
Engagement rate is directly calculated from likes, comments, shares, saves and impression which are also used as input features in the model. This creates a form of data leakage, where the model is essentially learning a mathematical relationship that already defines the target variable.

- Implications

- The high R2 score does not necessarily reflect the real-world predictive power.
- The model is not generalizing, but rather reconstructing the formula
- This inflates performance metrics artificially.

- Conclusion of Critical Analysis

The model demonstrates technical correctness but limited practical applicability unless

- Derived features are excluded
- New independent predictors are introduced

8. Key Insights

Based on exploratory analysis and modeling:

1. Engagement is strongly driven by interaction metrics
    - Likes, shares, and saves have a direct and dominant influence on engagement rate.

2. Non-linear relationships exist
    - Random Forest significantly outperformed Linear Regression, indicating complex interactions between variables.

3. Reach plays a balancing role
    - Higher reach does not always mean higher engagement rate, as it acts as the denominator

9. Recommendations

Based on the findings, the following recommendations are proposed:

1. Focus on Shareable and Savable Content
- Content that encourages users to share and save posts contributes significantly to higher engagement rates.

2. Optimize content strategy for Interaction
- Prioritize content formats that drive:

- Comments
- Likes
- Saves

3. Monitor Engagement Relative to Reach

- Instead of focusing solely on reach, evaluate performance using engagement rate to ensure content quality over quantity.

10. Future Work

- To improve the robustness and real-world applicability of the model:
- Remove or transform features directly used in the engagement formula
- Incorporate additional variables such as:
    - Post timing
    - Content type
    - Audience demographics
- Experiment with advanced models such as Gradient Boosting or Neural Networks
- Validate model performance on unseen or external datasets

11. Conclusion

This project successfully developed models to predict Instagram engagement rate with the Random Forest model achieving the highest performance.
However, a critical evaluation revealed that the model's performance is heavily influenced by data leakage, as the target variable is derived from the same features used for prediction. This highlights the importance of understanding feature relationships and validating model assumptions.
Despite this limitation, the analysis provides valuable insights into the drivers of engagement and offers actionable recommendations for improving content strategy.